06-07-2025

Agenda - Database IV

→ More SQL keywords / query.

→ Run new sql queries in pgAdmin 4.

→ working with cloud based postgres DB. → python

IN

let you check if a value matches any value in a list or subquery.

```
Select column_name from table_name where artist_id IN ('12', '13', '14');
```

```
a = [1,2,3]

for i in range(0,100,1):
    if i in a:
        print(i)
```

→ 1
2
3

wifi-subscription:

call → provider → agent → select → DSID

IN (DS1D, DS1D2)

```
SELECT *
FROM album
WHERE artist_id IN (1,2,3,4,5)
ORDER BY artist_id;
```

SQL Command within Query / SubQuery

→ use a query inside another query.

② → select artist_id from artist where name LIKE 'L%'

meaning: artist_id → start from L

→ 10 artist id

① → select * from album where artist_id IN (10 artist id)

→

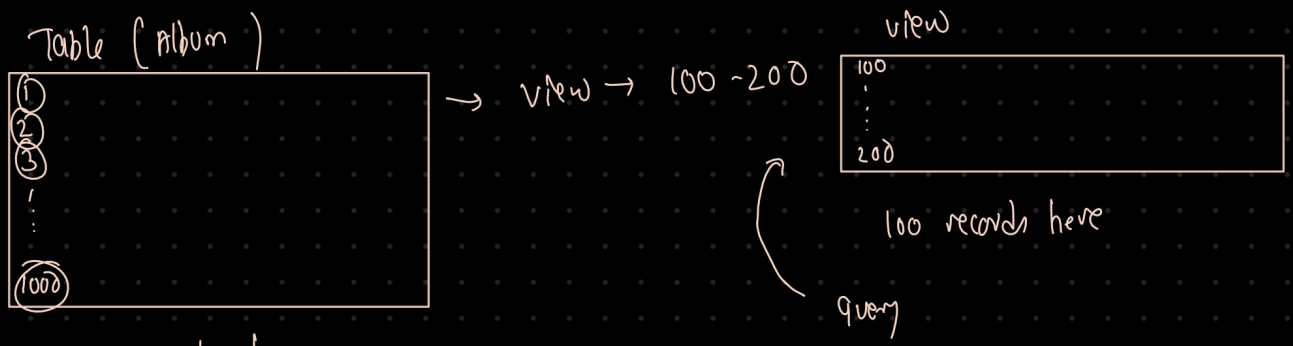
```
selec ↑ *
from album
where artist_id IN ( SECECT artist_id from artist where
                     Name Like 'L%' );
```

ILIKE → works on both case

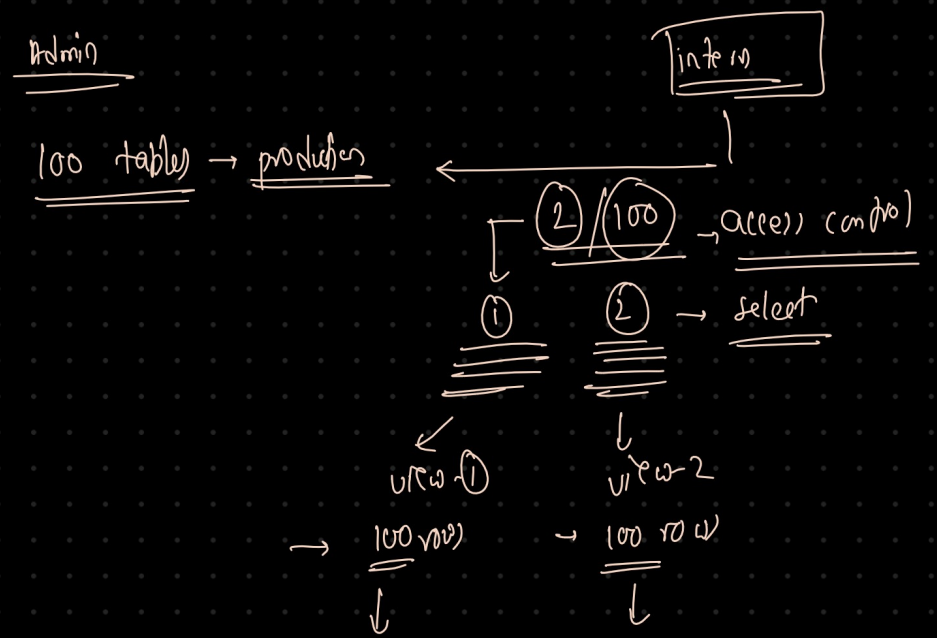
```
SELECT *
FROM album
WHERE artist_id IN (
   SELECT artist_id
   FROM artist
   WHERE name ILIKE 'L%'
)
ORDER BY artist_id;
```

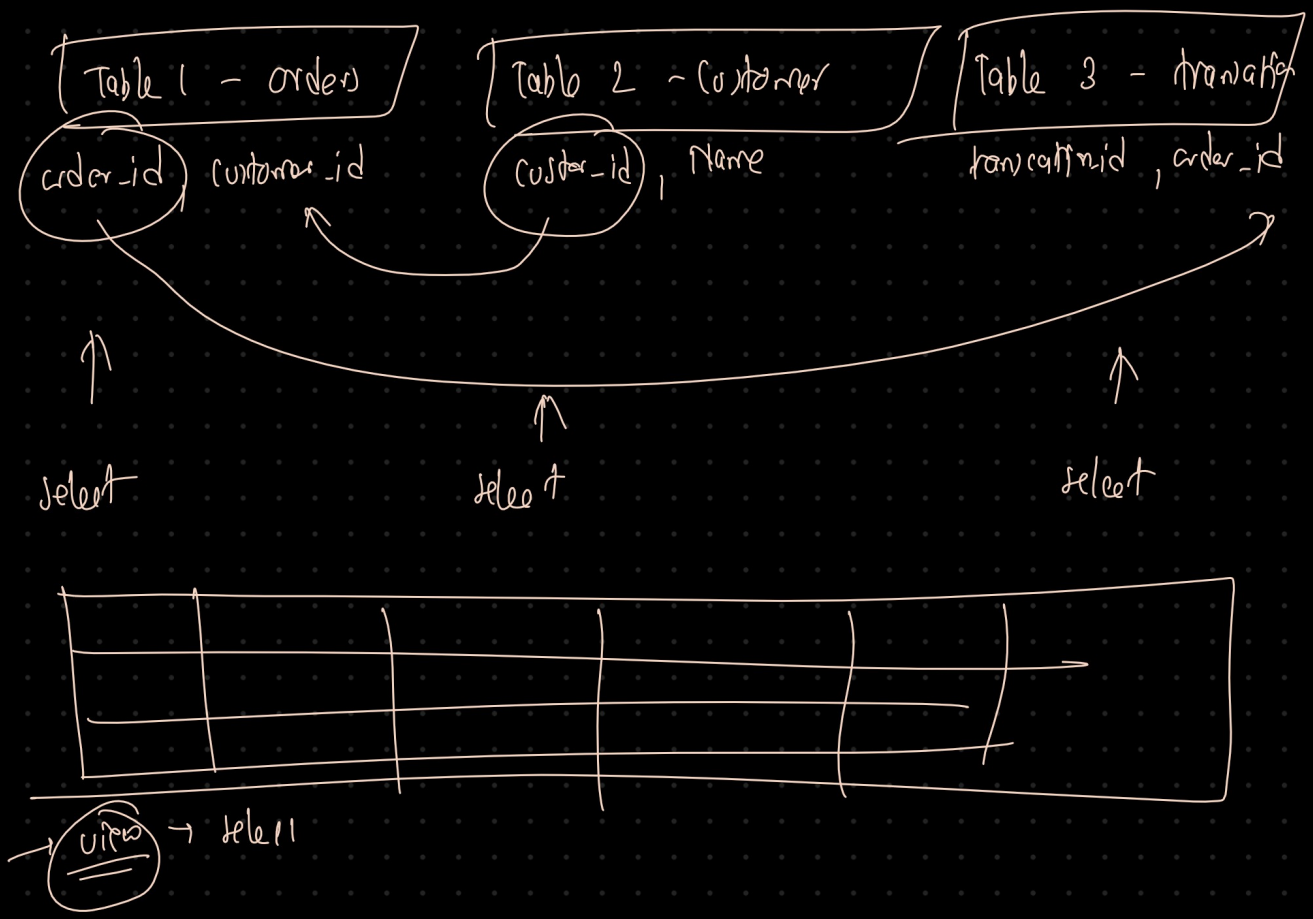
VIEW

- A view is a virtual table based on a query.
- It doesn't store data by itself, it shows live result whenever you query it.
- → Once we have the view (not a table), we can query it like a table.



create table ← view → view is stored in a database





View

```
CREATE VIEW big_invoices AS
SELECT invoice_id, customer_id, total
FROM invoice
WHERE total > 20;
```

} create view

```
DROP VIEW big_invoices;
```

} Drop view

Advantage :

- → Simplify frequent joins and filters
- → Improve readibility
- → Encapsulate business logic
- → control access to data / data security

Definition of view:

```
select * from pg_views
where viewname = 'big_invoices';
```

Advanle

||

Concatenation operator

```
Select 'Hello' || ' ' || 'world' AS greeting;
```

```
SELECT 'Hello' || ' ' || 'World' AS greeting;
```

name → artist
artist_id → artist

```
Select name, artist_id from artist;
```

```
Select artist.name , artist.artist_id from artist;
```

table_name.column_name (Syntax)

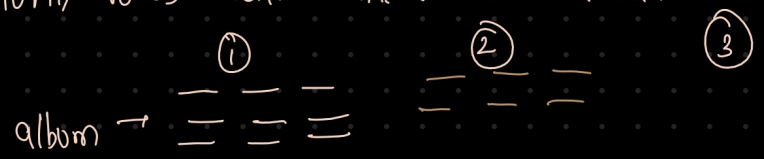
Joins:

Join combines rows from two or more table based on related column. (usually a foreign key)

- INNER JOIN
- LEFT JOIN (LEFT OUTER JOIN)
- RIGHT JOIN (RIGHT OUTER JOIN)
- FULL JOIN (FULL OUTER JOIN)
- CROSS JOIN

INNER JOIN

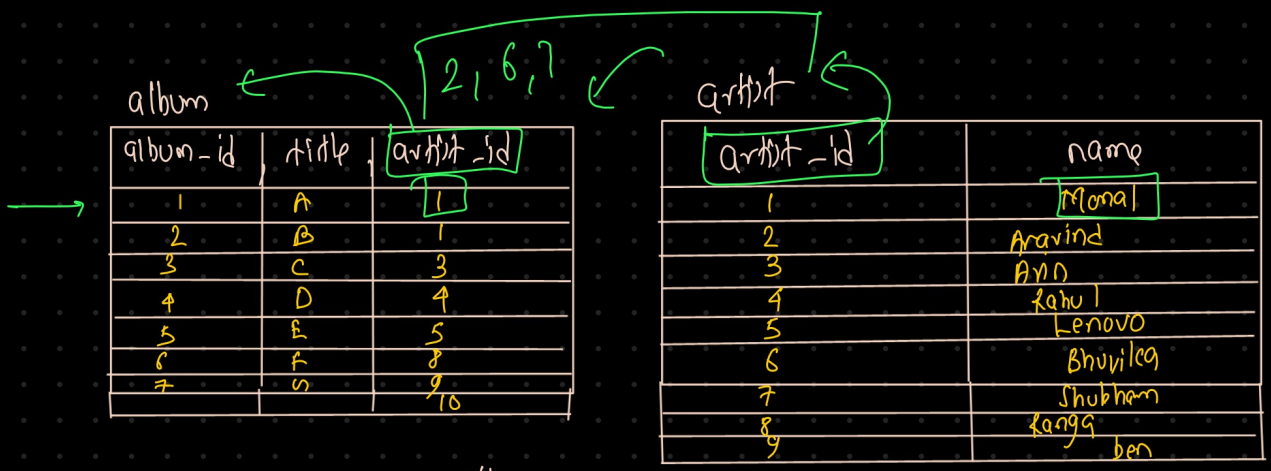
Return rows where there is a match in both tables.



artist → 1000 unique artist

album_id, title, name

p.s → list all albums with their artist name.



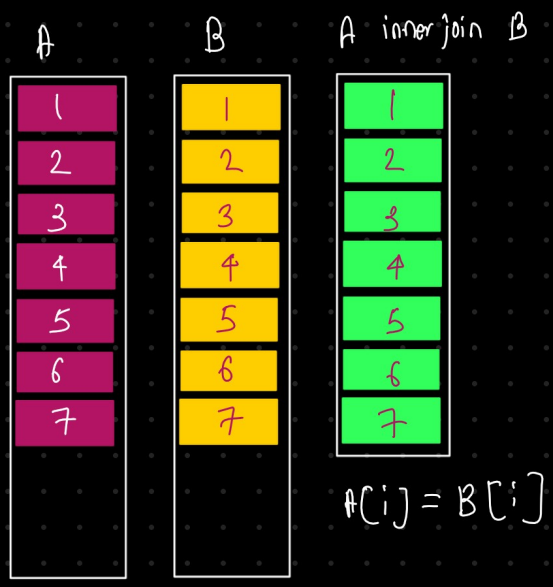
p.s → list all albums with their artist name.

```
select album.album_id, album.title, artist.name
```

get data only when both tables have that value

```
SELECT album.album_id, album.title, artist.name AS artist_name
FROM album
INNER JOIN artist
ON album.artist_id = artist.artist_id;
```

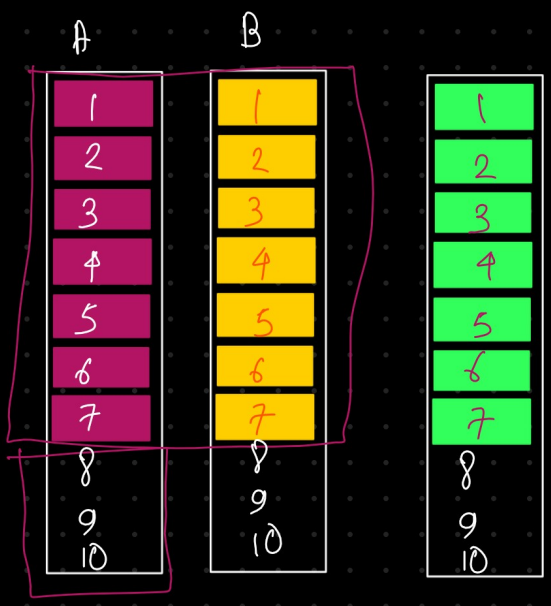
```
select al.album_id, al.title, art.name
from album al, artist art where
al.artist_id = art.artist_id;
```



LEFT JOIN

returns all rows from left table, plus matching rows from the right table.

If no match, the right side colums are Null



A

album_id	title	artist_id
1	A	1
2	B	1
3	C	2
4	D	2
5	E	10
6	F	10
7	G	8
8	H	8

B

artist_id	name
1	Monal
2	Aravind
3	Ann
4	Rahul
5	Lenovo
6	Bhuvilca
7	Shubham
8	Ranga
9	ben

Inner Join → A inner join B on A.artist_id = B.artist_id

→ 1, A, Monal
→ 2, B, Monal
→ 3, C, Aravind
→ 4, 0, Aravind
→ 7, G, ben
→

5 rows

LEFT JOIN

A Left Join B On A.artist_id = B.artist_id

→ 1, A, Monal
→ 2, B, Monal
→ 3, C, Aravind
→ 5, E, NULL
→ 4, 0, Aravind
→ 7, G, ben
→ 6, F, NULL
→ 8, H, NULL

5 + 3

RIGHT JOIN

B RIGHT JOIN A ON A.artist_id = B.artist_id

A

	album_id	title	artist_id
1	1	A	1
2 →	2	B	1
3	3	C	2
4	4	D	2
5	5	E	1
6	6	F	1
7	7	G	1
8	8	H	1

B

	artist_id	name
1	1	Mona
2	2	Aravind
3	3	Arin
4	4	Rahul
5	5	Lenovo
6	6	Bhuvilea
7	7	Shubham
8	8	Ranga
9	9	Ben

```
select artist.artist_id
       artist.name
       album.title
from
    album
    Right JOIN artist
on album.artist_id = artist.artist_id
```

1, mona, A
→ 1, mona, B
2, Aravind, C
→ 2, Aravind B
3, Arin, Null
4, Rahul, Null
5, lenovo, Null
6, Bhuvila, Null
7, Shubham, Null
8, Ranga, Null
9, Ben, G

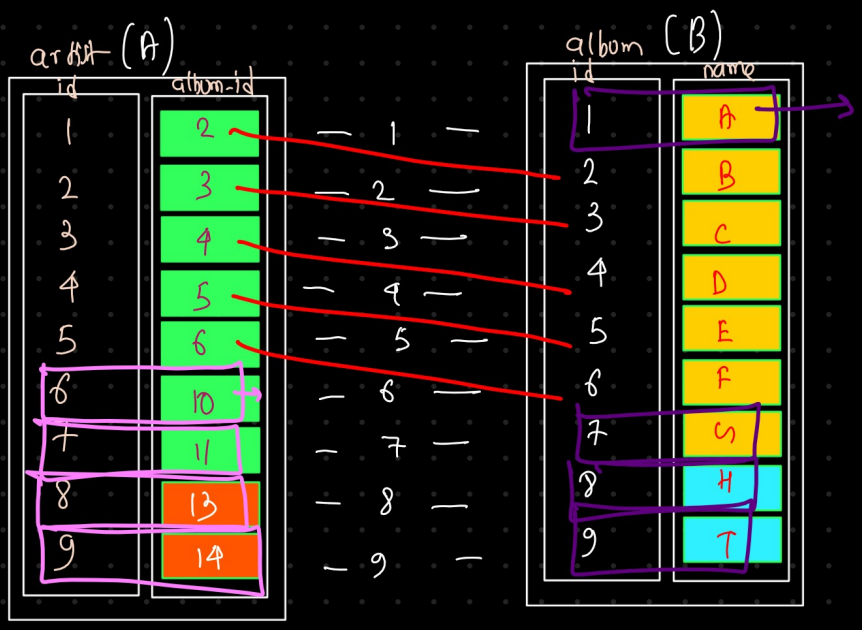
FULL OUTER JOIN

- return all rows from both table, matching rows where possible

→ Left A on B → 5 + 3

→ Right A on B → 5 + 4

→ 5 + 3 + 4 →



A-1 → B-2
A-2 — B-3
A-3 B-4
A-4 — B-5
A-5 — B-6

A-6 — NULL
A-7 → NULL
A-8 → NULL
A-9 — NULL

A

→ (common)

B

NULL — B-7
NULL — B-8
NULL — B-9
NULL — B-1

5 + 4 + 4 → 13

```sql
SELECT
  artist.artist_id,
  artist.name AS artist_name,
  album.title AS album_title
FROM
  album
RIGHT JOIN artist
  ON album.artist_id = artist.artist_id;

-- Show all artists and their albums,
-- including artists with no albums.
```

album Right Join Artist

↓

matching

↓

all

select artist.artist_id, artist.name, album.title

from

album